

Statistical Modelling & Predictive Analytics of the Irish Housing Market (2019–2024)

Objective

- Statistical modelling of Irish housing transactions
- Quantify market pressure, affordability, and risk
- Support forward-looking policy and investment decisions

Why Statistical Modelling?

- Regional disparities and demand-driven growth
- Rising affordability pressure and uneven volatility
- Enables predictive and causal insights

Decision-Making Impact

- Identify affordability stress and overheating across counties
- Support evidence-based housing supply and policy planning
- Assess risk-adjusted investment opportunities and market stability
- Anticipate future price growth and demand pressures

Key Model Insights

- Location and property condition drive prices
- Sales growth aligns with rapid price inflation
- High-demand counties face lowest affordability
- Second-hand properties show higher volatility

Modelling Approach

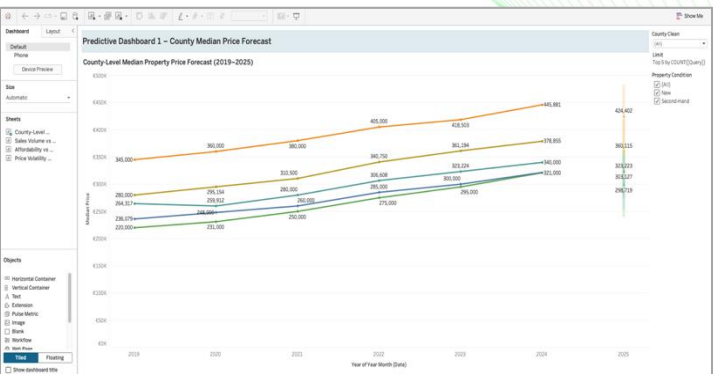
- Time-series forecasting (ETS)
- Linear regression modelling
- Affordability stress analysis
- Price volatility metrics (IQR)

Data & Methodology

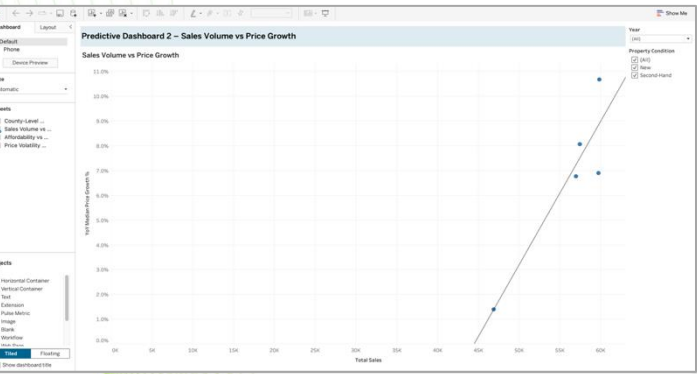
- PSRA Property Price Register (2019–2024)
- County-level, transaction-level data
- EDA, regression, forecasting, volatility analysis

Predictive Dashboards Overview

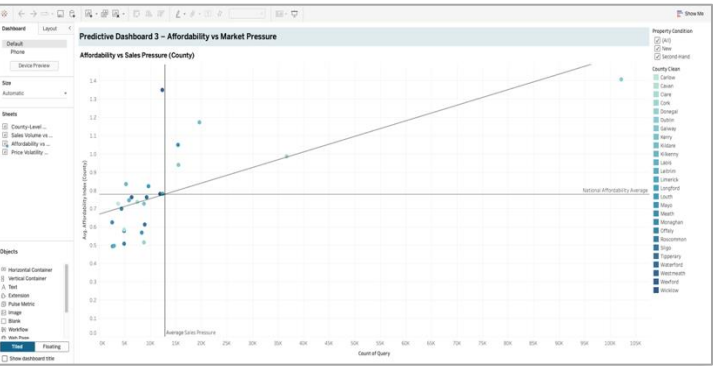
County Median Price Forecasts



Sales Volume vs Price Growth Analysis



Affordability Stress Classification



Price Volatility Risk Profiling

